is much weaker than the nuclear force, so that in this case, practically, only the latter is operative. However, due to its long range, the electric force continues to be perceptible long after the nuclear force has ceased to be operative. If we draw a diagram of the potential energy of a proton at various distances from another proton, it will look more or less like the one reproduced in Figure 16. In fact, up to a distance of the order of 5×10^{-13} cm., the picture is practically identical with the one shown in Figure 15. From that point on, however, the potential energy does not approach 0 asymptotically, but passes through 0, rises to positive magnitudes, and only then does it drop asymptotically toward 0. Between two protons there exists what we call a *potential barrier,* a specimen of which will be discussed repeatedly later on.

According to all this, one might surmise that there exists a state in which two individual protons are bound to each other, namely, when their distances from each other are so small that the nuclear force of attraction overcomes the electric force of repulsion. But this is probably never the case. As pointed out before, two particles bound to each other vibrate constantly in relation to each other, even in their normal or ground state, the state of least energy. This 'zero-point vibration' is probably so powerful that it makes it impossible for any permanent bond to exist between two individual protons. But the attraction between protons is certain to play an important part in the more complex nuclei.

So now we have obtained our first overall view of nuclear forces. The most important of these is the force of attraction between neutron and proton. There is, furthermore, a force of a similar order of magnitude acting between two protons or two neutrons. The operative range of these nuclear forces diminishes rapidly with the distance, and in this respect they resemble the chemical valency forces, which likewise possess a very short range only.

II. THE NUCLEAR FORCES AS EXCHANGE FORCES

Let us now continue to formulate our queries in the same way as we did when dealing with the electric forces. Thus our first question will be: Does there exist any analogy permitting us to link nuclear forces with particles in a way similar to that in which we link electric forces with photons? With this object in view, we must once again study the chart appearing on page 60. The building blocks of the extranuclear structure of the atom are the electrons which are bound to the nucleus by the electric field. The electric field, in turn, is linked with the photons emitted by the atom when certain changes occur in this extranuclear structure. The building blocks of the nucleus are the neutrons and protons, which are held together by the nuclear field, while in